Supplementary Information

Document caption: Detailed methods, supplementary analyses, and results including p-values. // læsevjeldning i appendix // tilføj significans niveau på figurer (caption)

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List of Figures

1 Current climate conditions

Data is available 2007 – 2022. 2008 is the first whole year of temperature data. Data available from https://data.g-e-m.dk (GEM 2026e, 2026c).

1.1 Temperature

Monthly mean temperature

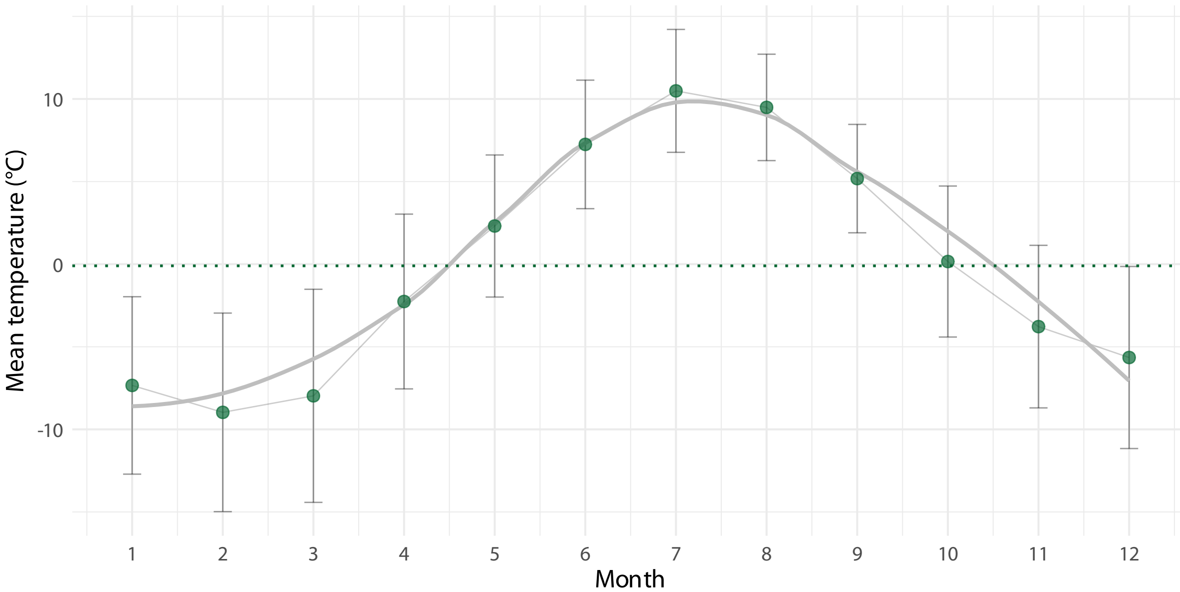

Figure 1: Mean temperature per month. Mean air temperature (°C) per month, based on data from 2007 to 2022. The dashed line represent yearly mean of -0.1°C.

The mean annual temperature is -0.1 °C. The warmest months are July (10.5 °C), August (9.5 °C), June (7.2 °C). The coldest months are January (-7.3 °C), March (-8 °C), February (-9 °C).

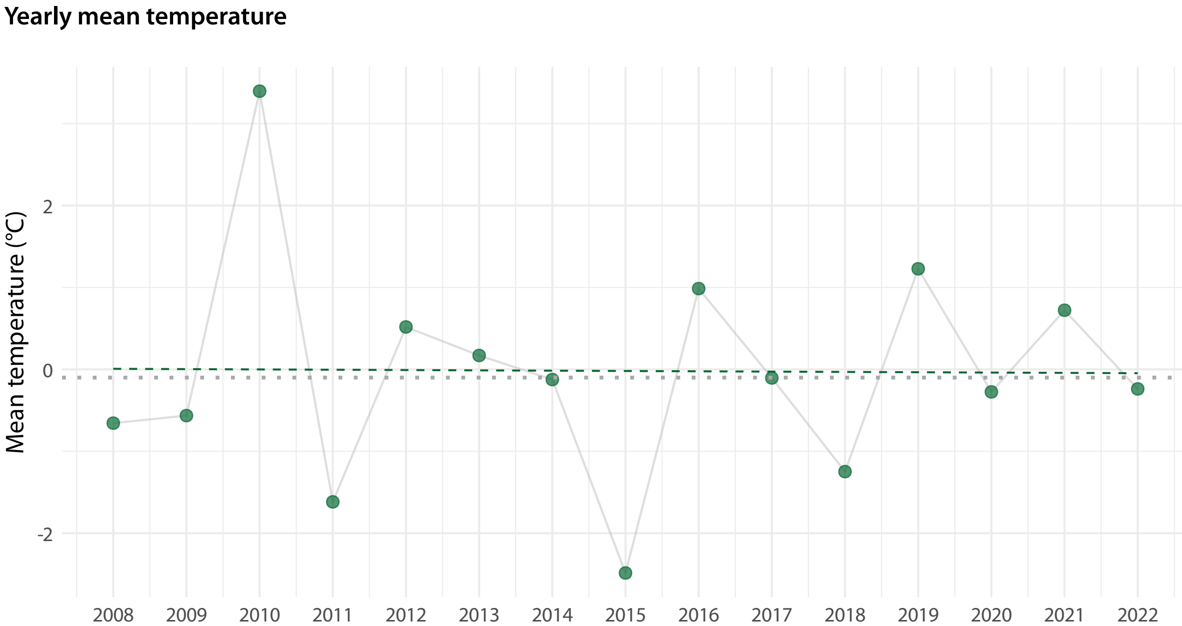

Figure 2: Mean annual air temperature per year (°C) from 2008 to 2022 in Kangerluassunguaq. Data from 2007 is excluded because data is only from October, November and December. Gray dotted line indicate over all mean of -0.1003°C. Dashed line is model prediction.

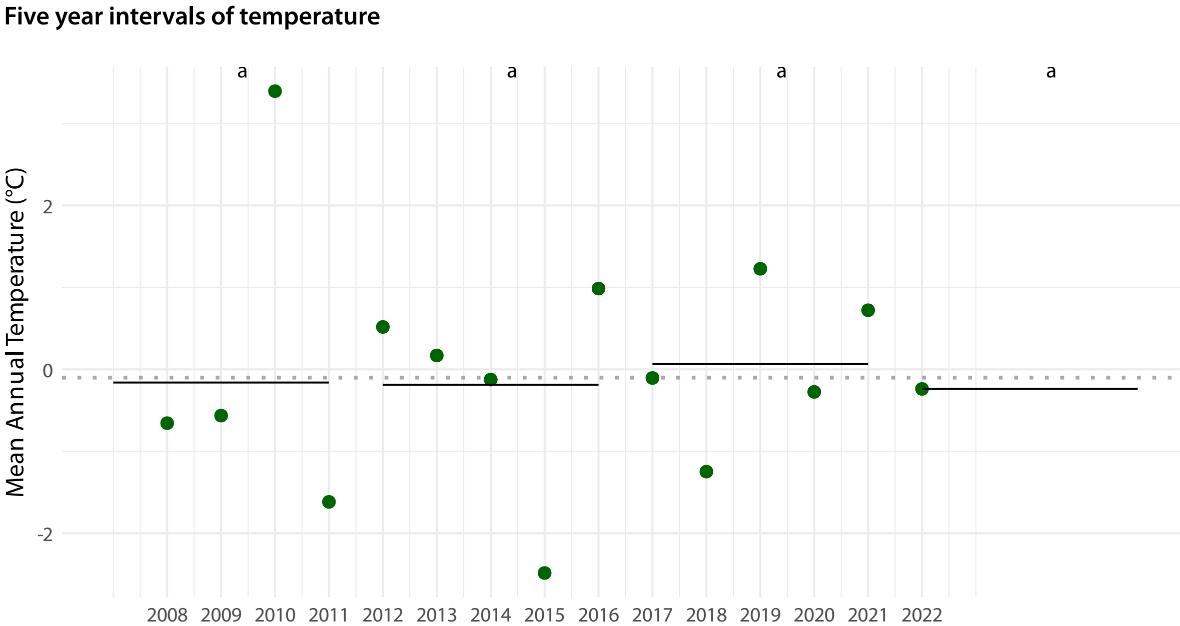

Figure 3: Five year intervals in temperature means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

1.2 Precipitation

Data available from https://data.g-e-m.dk (GEM 2026c).

The mean annual precipitation is 886.15 mm.

The months with the most precipitation are September (138.64 mm), October (99.38 mm), August (98.28 mm), which account for 336.3 mm or 38 % of the yearly precipitation.

The months with the least precipitation are May (51.51 mm), February (37.62 mm), March (36.44 mm).

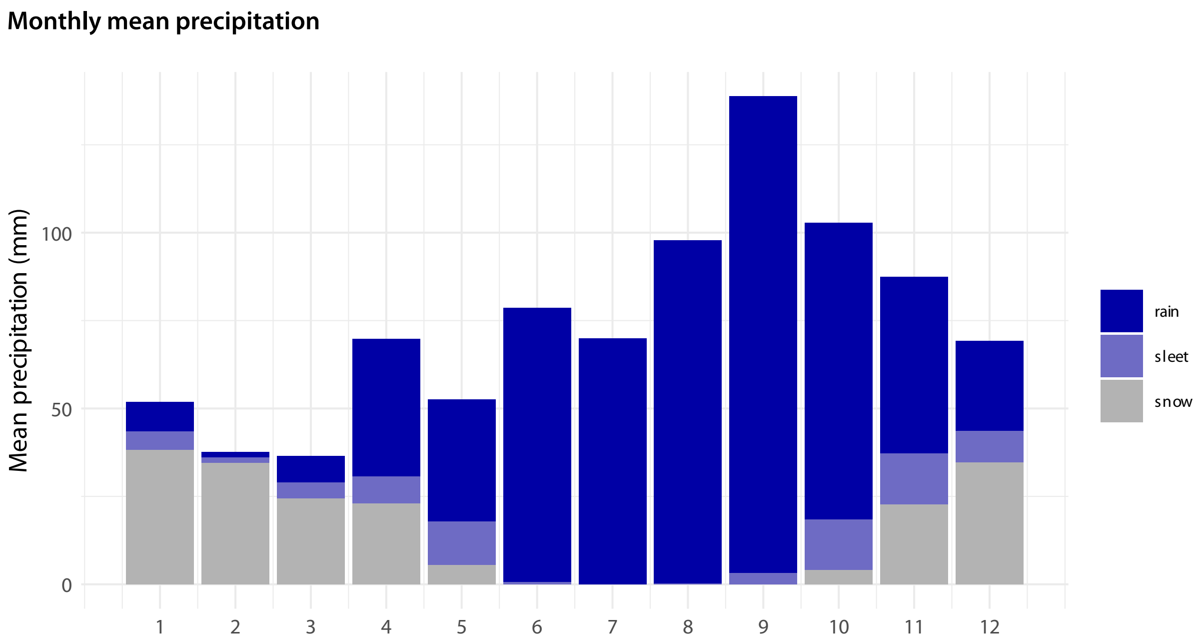

Figure 4: Mean precipitation type per month. Mean monthly distribution of types of precipitation (mm). Totals based on data from 2007 to 2024 (requires join with temperature data which it only available till -Inf). Sleet is defined as precipitation that fell when temperatures were between -1°C and 1°C.

Yearly mean precipitation

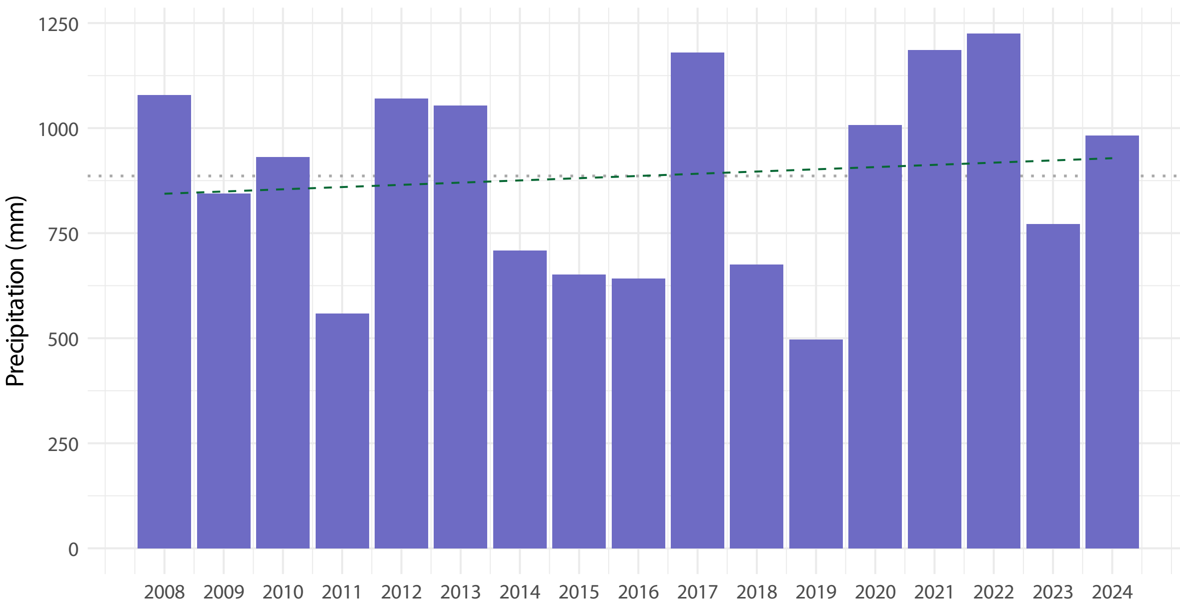

Figure 5: Mean yearly precipitation (mm) from 2008 to 2024. Data from 2007 is excluded because it was only from May - December. Blue dashed line indicate over all mean of 886.15 mm. Dashed line represent model prediction.

Five year interval precipitation

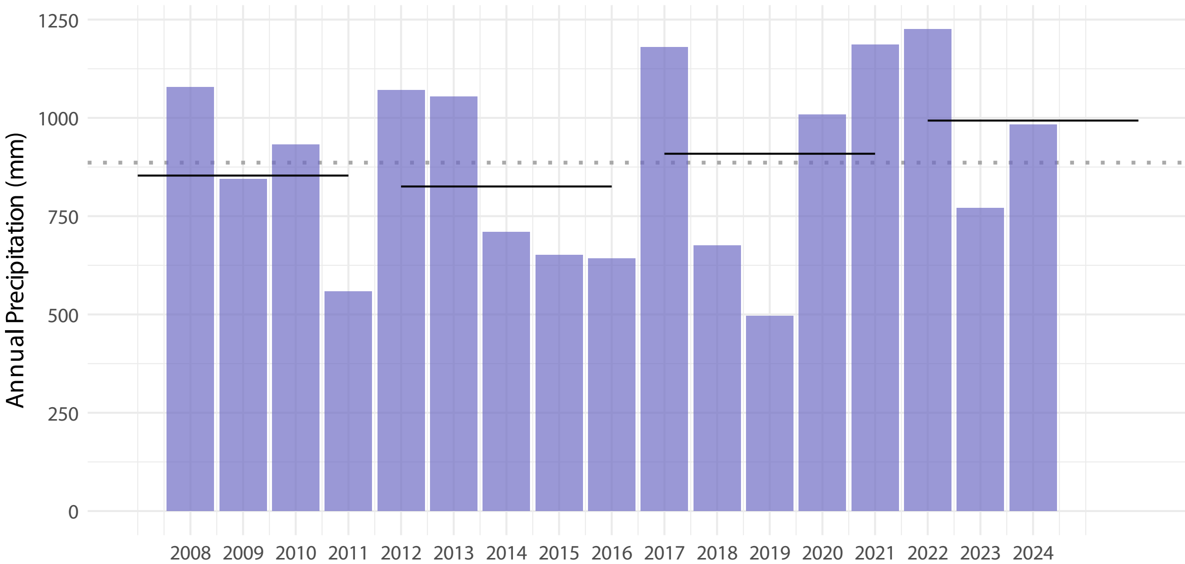

Figure 6: Five year intervals in precipitation means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

1.3 Radiation

Data available from https://data.g-e-m.dk (GEM 2026d)

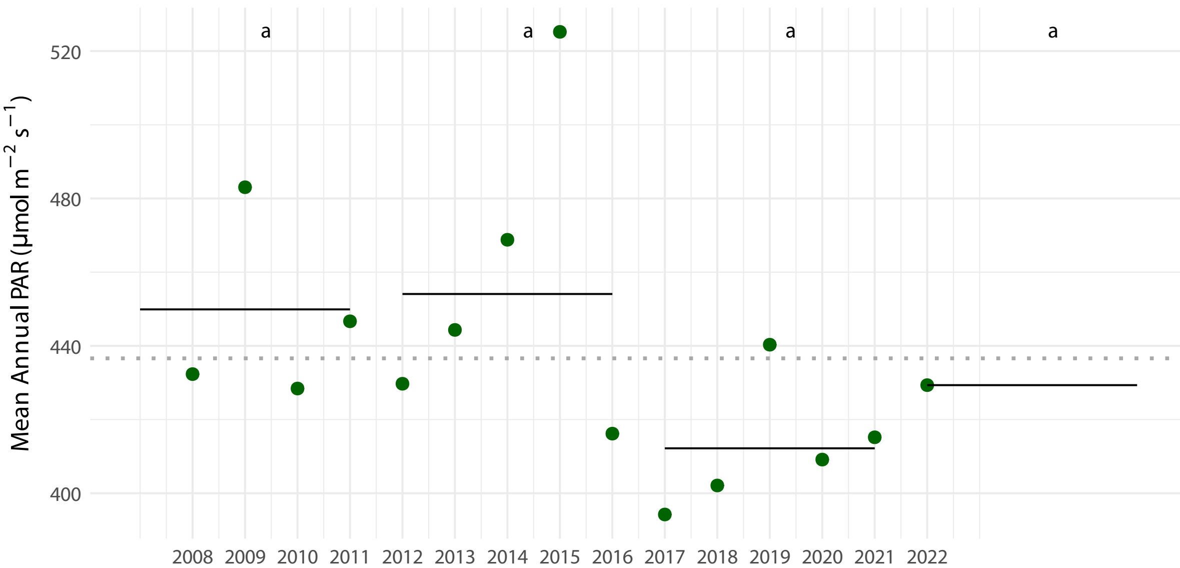

Figure 7: Five year intervals in PAR means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

2 Historic climate data

2.1 Temperature

Data available from Cappelen (2021).

Station 4250 is Nuuk, and element 101 is precipitation.

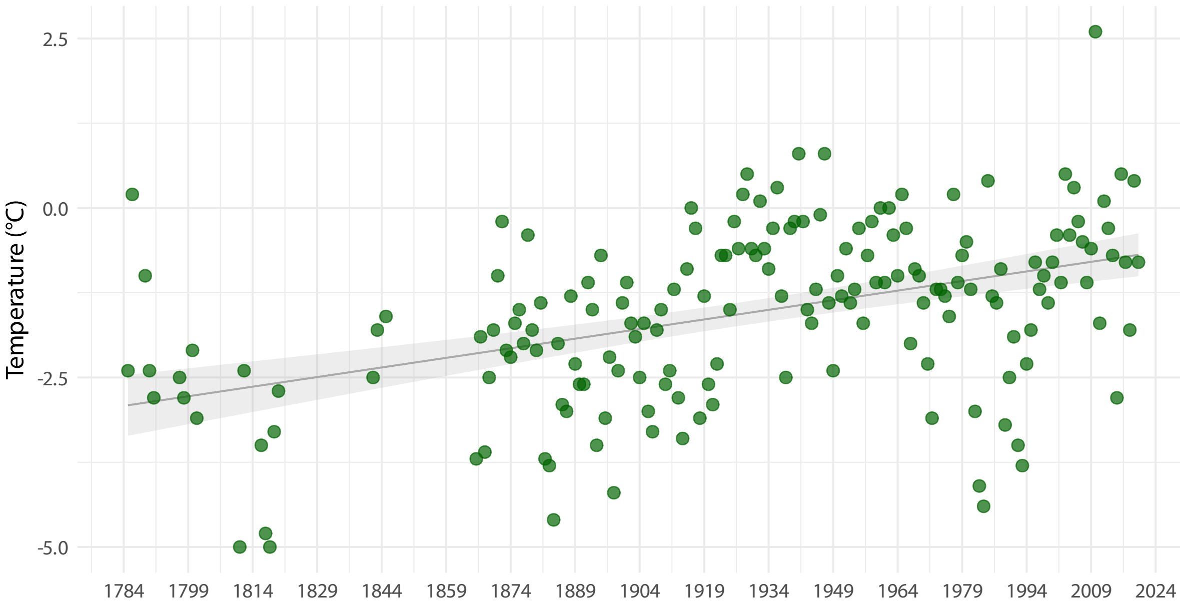

Figure 8: Historic yearly mean air temperature (°C) in Nuuk, from 1785 to 2020. Yearly mean of -1.6°C. Yearly mean ranges from -5°C to 2.6°C. Modelled mean temperature has increase from -2.92°C in 1784 to -0.65°C in 2024 (increase of 2.27°C or 0.095°C/decade.)

3 Statistical analysis

3.1 Diversity

Data available from https://data.g-e-m.dk (GEM 2026a).

Data has been filtered to remove sections with only one remaining plot and subsections in saltmarsh, due to too few replicates in this community type. Final calculates included 134 subsections (full transect consists of 162 subsection).

Richness

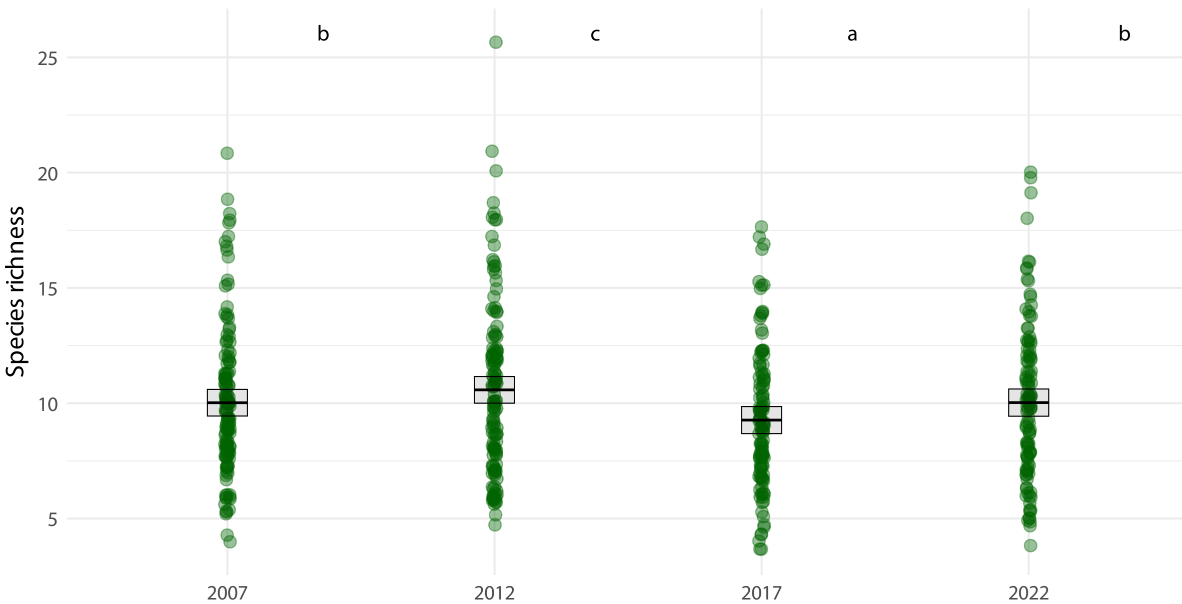

Figure 9: Species richness, number of species pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

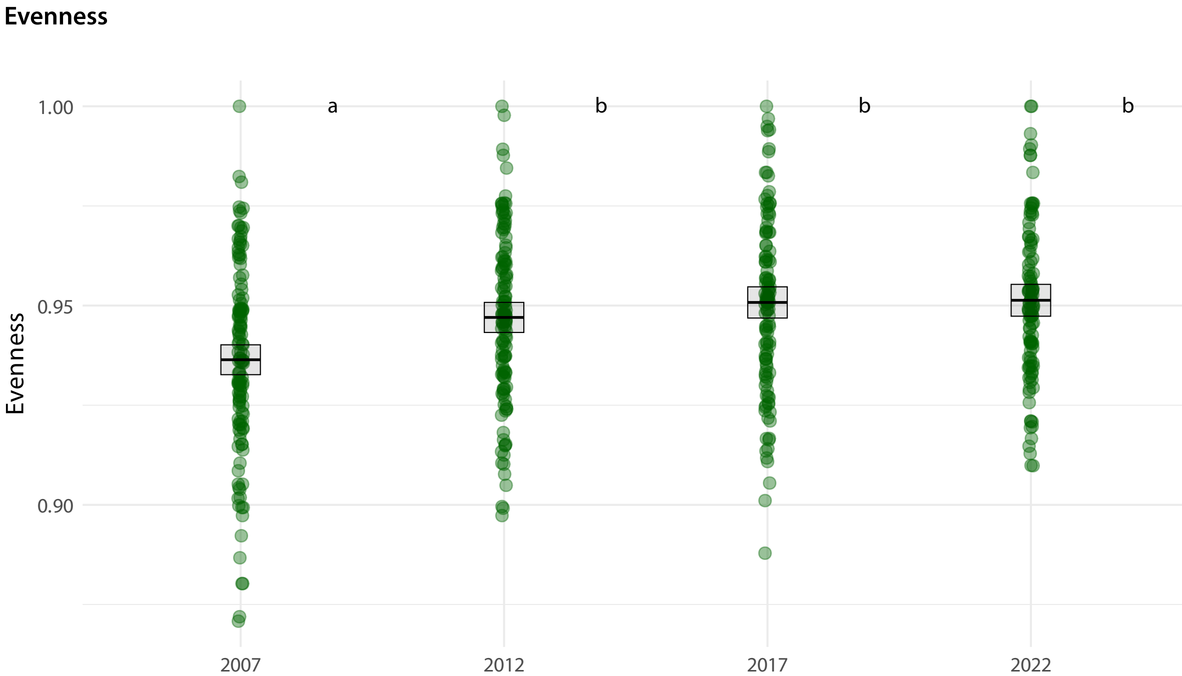

Figure 10: Evenness pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Shannon

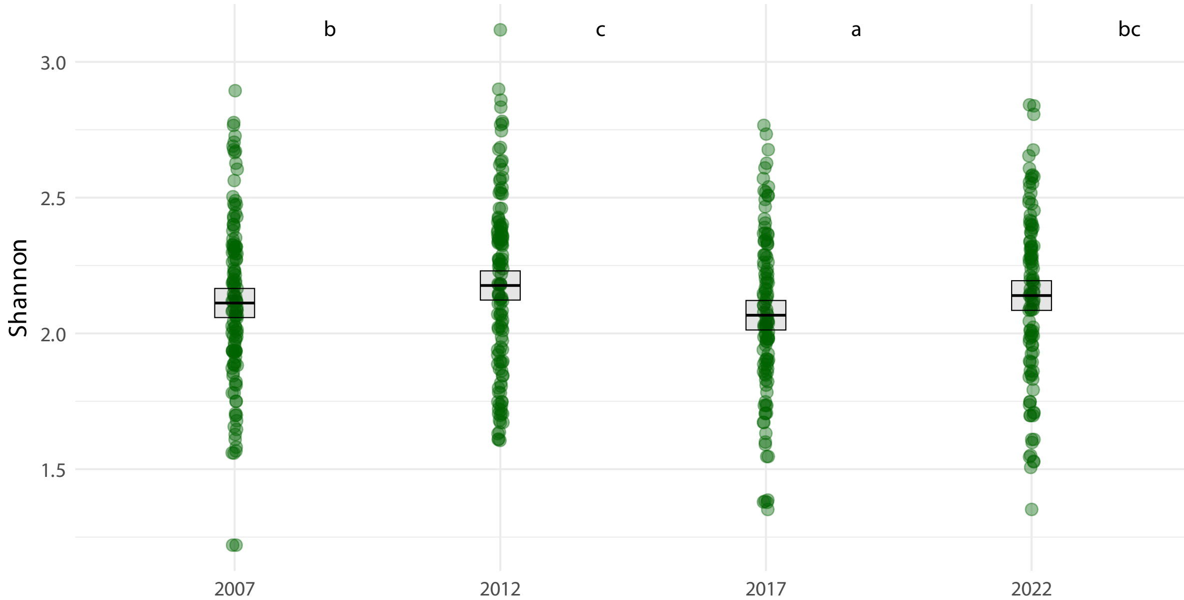

Figure 11: Shannon diversity pr. year calculate pr subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Turnover

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(`geom_crossbar()`).
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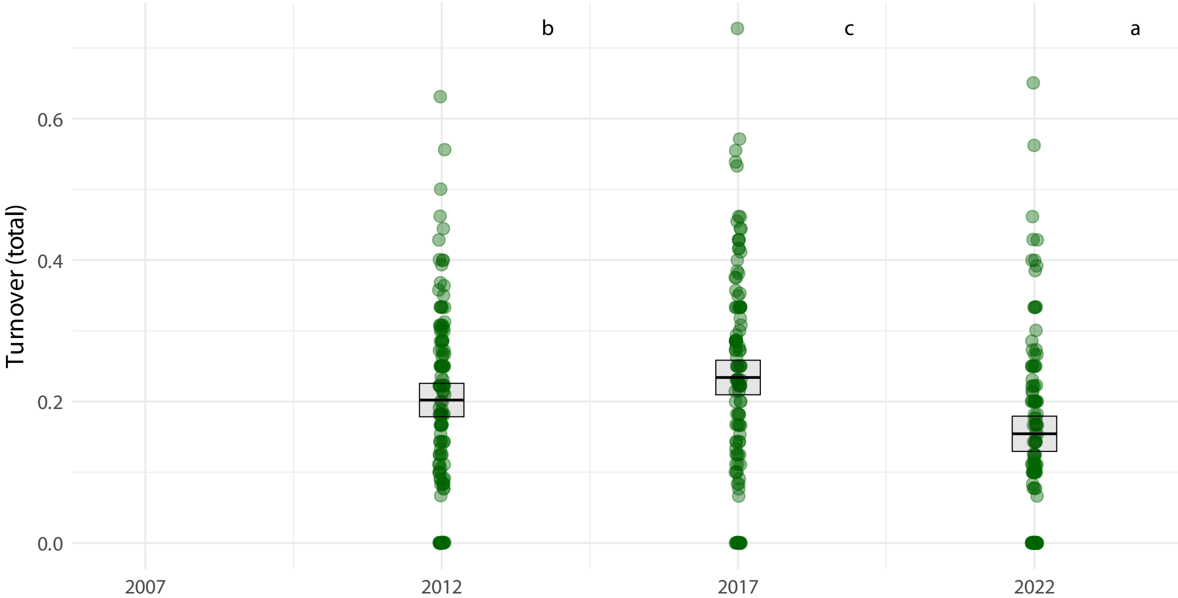

Figure 12: Turnover pr. year calculate pr subsection, but only pr present all survey years. A totalt of 131 subsections maintain more than 1 plots for all 4 surveys. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Plots combined

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(`geom_crossbar()`).
Removed 1 row containing missing values or values outside the scale range
(`geom_crossbar()`).
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[1] "/Users/ibdj/Library/CloudStorage/OneDrive-Aarhusuniversitet/MappingPlants/01 Vegetation
```

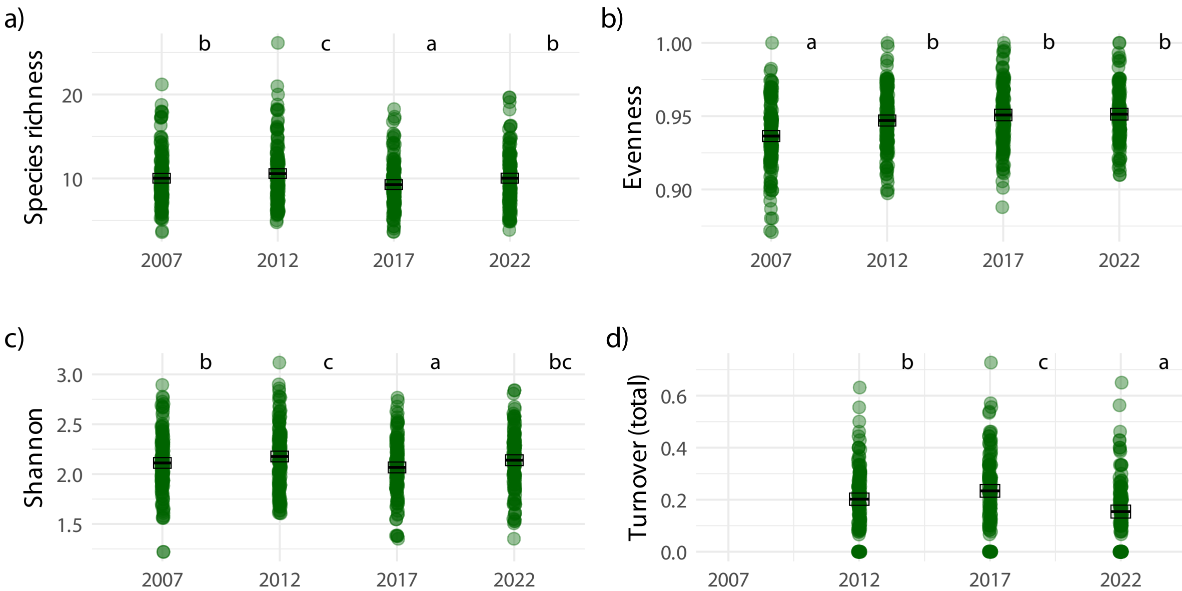

Figure 13: Diversity metrices, a) Shannon diversity, b) Species richness, c) Evenness, c) Species turnover. Shared letters indicate non-significant differences among estimated means according to compact letter display.

3.2 Community composition

NMDS

Community composition data ready for analysis consists of 134 subsections from 4 years of surveys. 99 species have been observed across all surveys.

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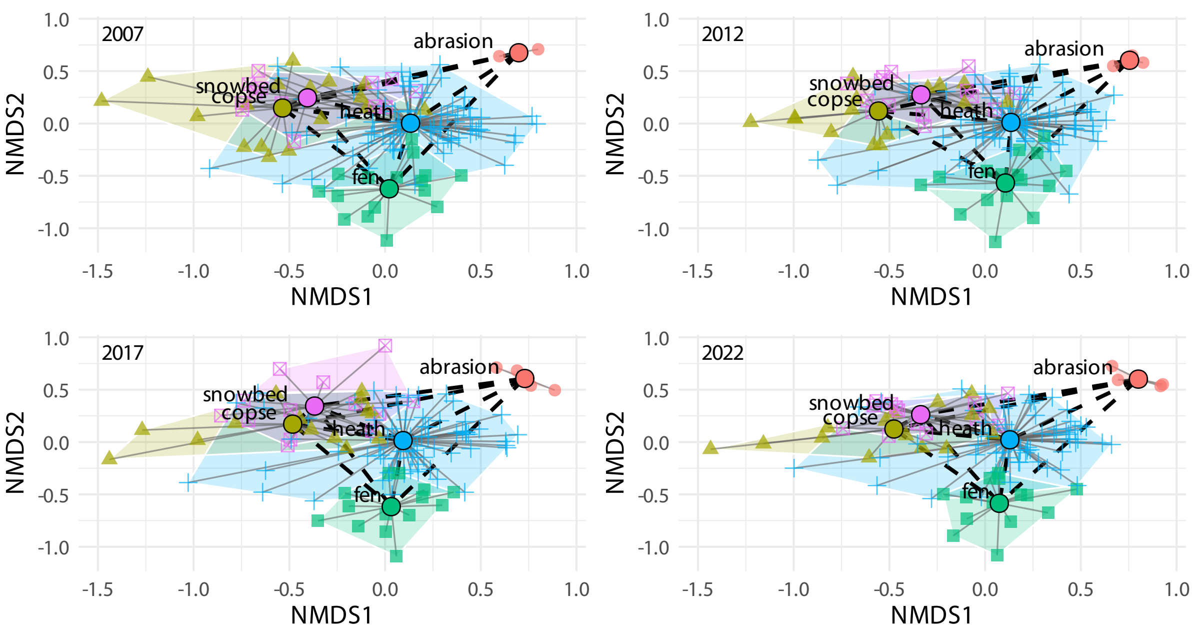

Figure 14: Plant community dynamics. Non-metric multidimensional scaling (NMDS) ordination of vegetation composition across years and vegetation types. Each panel represents subset of years of the full NMDS including all data to keep NMDS distances comparable. Grey lines connect plots within vegetation types, the within-type centroid distances. Black dashed lines indicate distances between vegetation-type centroids.

Between vegetation composition

```
Model: dist ~ factor(year) + (1 | veg_pair), fitted by REML
```

```
N observations: 40 | N groups (veg_pair): 10
```

Table 1: Fixed effects: centroid distance ~ survey year (factor)

term	estimate	std.error	statistic
(Intercept)	0.886	0.119	7.462
Year 2012	-0.014	0.017	-0.780
Year 2017	-0.018	0.017	-1.024
Year 2022	-0.025	0.017	-1.448

Table 2: Estimated mean centroid distances per survey

year	emmean	SE	lower.CL	upper.CL
2007	0.886	0.119	0.618	1.154
2012	0.873	0.119	0.605	1.140
2017	0.868	0.119	0.600	1.136
2022	0.861	0.119	0.593	1.129

Table 3: Pairwise contrasts between surveys (Sidak-adjusted)

contrast	estimate	SE	t.ratio	p.value
year2007 - year2012	0.014	0.017	0.780	0.970
year2007 - year2017	0.018	0.017	1.024	0.897
year2007 - year2022	0.025	0.017	1.448	0.647
year2012 - year2017	0.004	0.017	0.244	1.000
year2012 - year2022	0.012	0.017	0.668	0.986
year2017 - year2022	0.007	0.017	0.424	0.999

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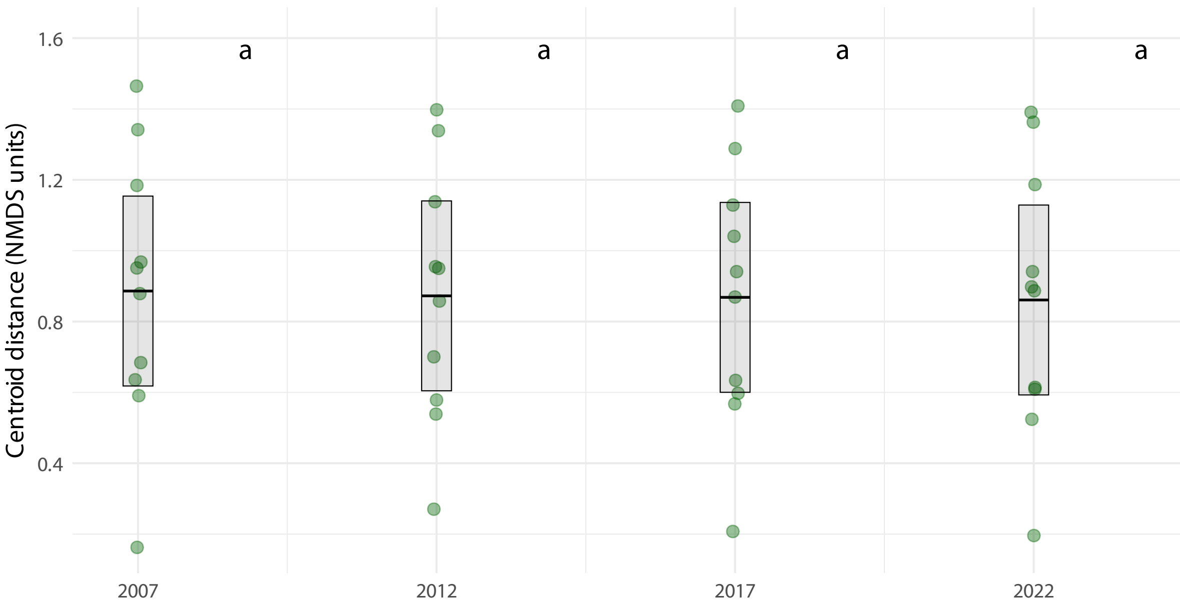

Figure 15: Between plant community dynamics. Estimated mean distance between vegetation-type centroids within each survey year based on linear mixed-effects models. Black lines are estimated means (± SE), and grey dots represent raw subsection-level distances (NMDS units). Shared letters indicate non-significant differences among estimated means according to compact letter display. (Figure 5)

Within vegetation composition

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[1] "/Users/ibdj/Library/CloudStorage/OneDrive-Aarhusuniversitet/MappingPlants/01 Vegetation
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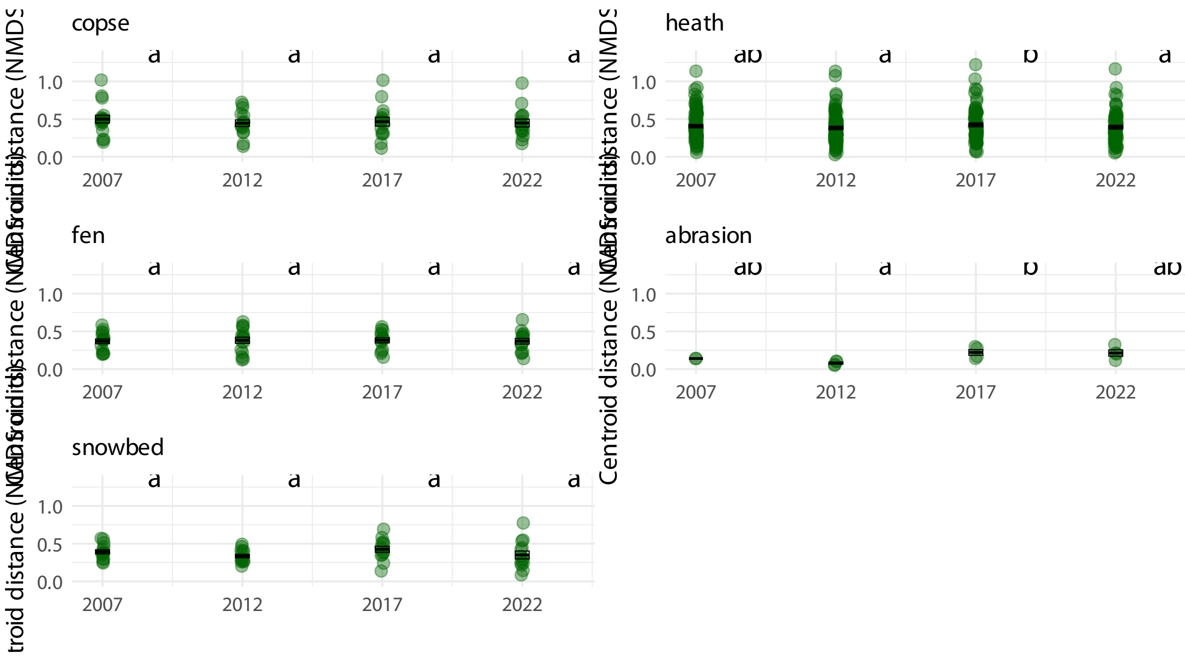

Figure 16: Within community dynamics. Estimated mean centroids-subsection distances pr community each survey year based on linear mixed-effects models. Black lines are estimated means (± SE), and grey dots represent raw distances (NMDS units). Shared letters indicate non-significant differences among estimated means according to compact letter display. (Figure 6)

3.3 Plant functional types

Functional groups

Table 4: Fixed effects per functional type

func_type	effect	term	estimate	std.error	statistic
forb	fixed	(Intercept)	0.268	0.029	9.224
forb	fixed	factor(year)2012	0.082	0.033	2.502
forb	fixed	factor(year)2017	0.004	0.034	0.114
forb	fixed	factor(year)2022	0.038	0.034	1.098
graminoid	fixed	(Intercept)	0.469	0.028	16.860
graminoid	fixed	factor(year)2012	0.078	0.038	2.022
graminoid	fixed	factor(year)2017	0.054	0.040	1.367
graminoid	fixed	factor(year)2022	0.080	0.039	2.025
lichen	fixed	(Intercept)	0.491	0.035	14.063
lichen	fixed	factor(year)2012	0.141	0.021	6.754
lichen	fixed	factor(year)2017	-0.047	0.021	-2.216
lichen	fixed	factor(year)2022	0.050	0.022	2.317
shrub	fixed	(Intercept)	0.639	0.032	19.938
shrub	fixed	factor(year)2012	0.065	0.046	1.404
shrub	fixed	factor(year)2017	0.052	0.047	1.098
shrub	fixed	factor(year)2022	0.068	0.048	1.441
bryophyte	fixed	(Intercept)	0.886	0.016	55.412
bryophyte	fixed	factor(year)2012	0.062	0.016	3.940
bryophyte	fixed	factor(year)2017	-0.009	0.016	-0.597
bryophyte	fixed	factor(year)2022	0.062	0.016	3.870

Table 5: Estimated marginal means per functional type and survey year

func_type	year	emmean	SE	df	lower.CL	upper.CL
forb	2007	0.268	0.029	456.607	0.211	0.326
forb	2012	0.351	0.028	422.755	0.295	0.407
forb	2017	0.272	0.030	491.535	0.213	0.331
forb	2022	0.306	0.030	493.655	0.247	0.365
graminoid	2007	0.469	0.028	907.475	0.415	0.524
graminoid	2012	0.547	0.028	929.536	0.491	0.603
graminoid	2017	0.523	0.030	1007.579	0.465	0.582
graminoid	2022	0.549	0.030	958.333	0.491	0.608
lichen	2007	0.491	0.035	182.008	0.422	0.560

func_type	year	emmean	SE	df	lower.CL	upper.CL
lichen	2012	0.632	0.035	182.939	0.563	0.701
lichen	2017	0.444	0.035	186.170	0.375	0.513
lichen	2022	0.541	0.035	190.104	0.471	0.611
shrub	2007	0.639	0.032	1080.340	0.576	0.702
shrub	2012	0.703	0.033	1143.720	0.638	0.768
shrub	2017	0.691	0.035	1201.696	0.622	0.760
shrub	2022	0.707	0.035	1161.022	0.638	0.776
bryophyte	2007	0.886	0.016	298.574	0.854	0.917
bryophyte	2012	0.947	0.016	300.019	0.916	0.979
bryophyte	2017	0.876	0.016	308.898	0.844	0.908
bryophyte	2022	0.948	0.016	320.224	0.916	0.980

Table 6: Pairwise contrasts per functional type (Holm-adjusted)

func_type	contrast	estimate	SE	df	t.ratio	p.value
forb	year2007 - year2012	-0.082	0.033	1055.408	-2.502	0.075
forb	year2007 - year2017	-0.004	0.034	1063.463	-0.114	0.909
forb	year2007 - year2022	-0.038	0.034	1071.277	-1.098	0.818
forb	year2012 - year2017	0.078	0.034	1057.885	2.329	0.100
forb	year2012 - year2022	0.044	0.034	1066.119	1.314	0.757
forb	year2017 - year2022	-0.034	0.035	1050.717	-0.973	0.818
graminoid	year2007 - year2012	-0.078	0.038	1471.251	-2.021	0.259
graminoid	year2007 - year2017	-0.054	0.040	1487.776	-1.367	0.688
graminoid	year2007 - year2022	-0.080	0.039	1498.188	-2.024	0.259
graminoid	year2012 - year2017	0.024	0.040	1476.967	0.590	1.000
graminoid	year2012 - year2022	-0.002	0.040	1484.852	-0.056	1.000
graminoid	year2017 - year2022	-0.026	0.041	1466.995	-0.631	1.000
lichen	year2007 - year2012	-0.141	0.021	391.177	-6.754	0.000
lichen	year2007 - year2017	0.047	0.021	391.832	2.216	0.042
lichen	year2007 - year2022	-0.050	0.022	392.251	-2.316	0.042
lichen	year2012 - year2017	0.188	0.021	390.979	8.893	0.000
lichen	year2012 - year2022	0.091	0.021	391.153	4.256	0.000
lichen	year2017 - year2022	-0.097	0.022	390.510	-4.491	0.000
shrub	year2007 - year2012	-0.065	0.046	1711.909	-1.403	0.900
shrub	year2007 - year2017	-0.052	0.047	1742.272	-1.098	1.000
shrub	year2007 - year2022	-0.068	0.048	1761.154	-1.440	0.900
shrub	year2012 - year2017	0.013	0.048	1729.213	0.260	1.000
shrub	year2012 - year2022	-0.004	0.048	1746.571	-0.079	1.000
shrub	year2017 - year2022	-0.016	0.050	1711.778	-0.330	1.000
bryophyte	year2007 - year2012	-0.062	0.016	392.994	-3.940	0.000

func_type	contrast	estimate	SE	df	t.ratio	p.value
bryophyte	year2007 - year2017	0.009	0.016	394.921	0.596	1.000
bryophyte	year2007 - year2022	-0.062	0.016	396.342	-3.870	0.000
bryophyte	year2012 - year2017	0.071	0.016	393.119	4.489	0.000
bryophyte	year2012 - year2022	-0.001	0.016	393.796	-0.038	1.000
bryophyte	year2017 - year2022	-0.072	0.016	391.918	-4.440	0.000

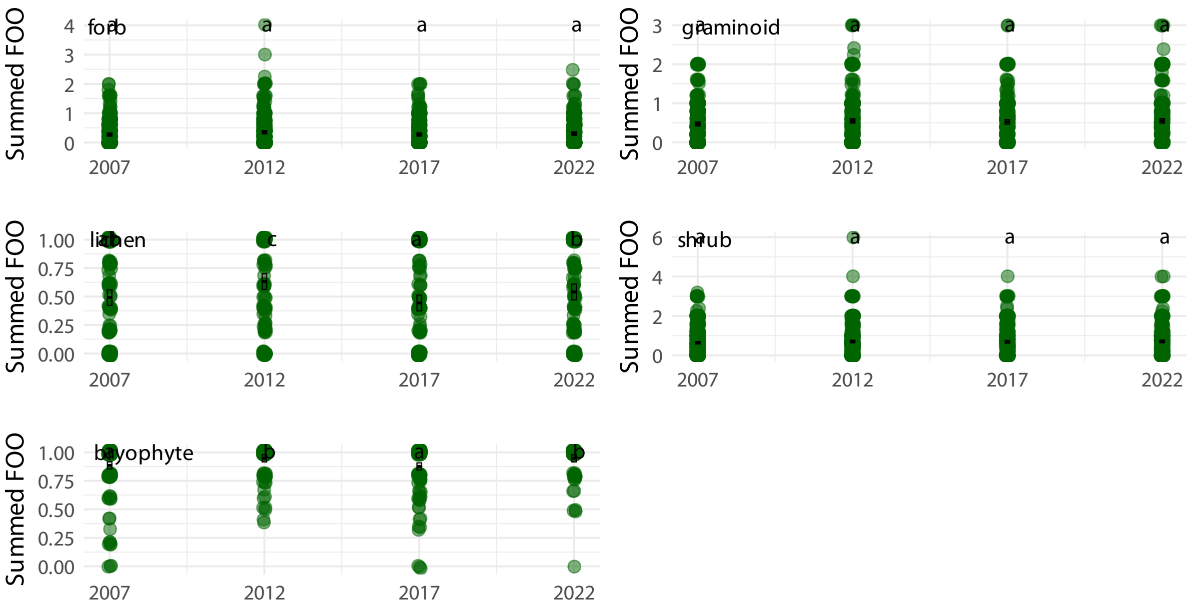

Figure 17: Plant functional group dynamics. Summed frequency of occurrence (FOO) for plant functional groups. Shared letters indicate non-significant differences among estimated means according to compact letter display. (Figure 7)

Individual species

A total of 97 species were analysed. 18 of them had any significant differences in means between survey years. Plots of the frequency of occurrence for these 18 species are rendered below.

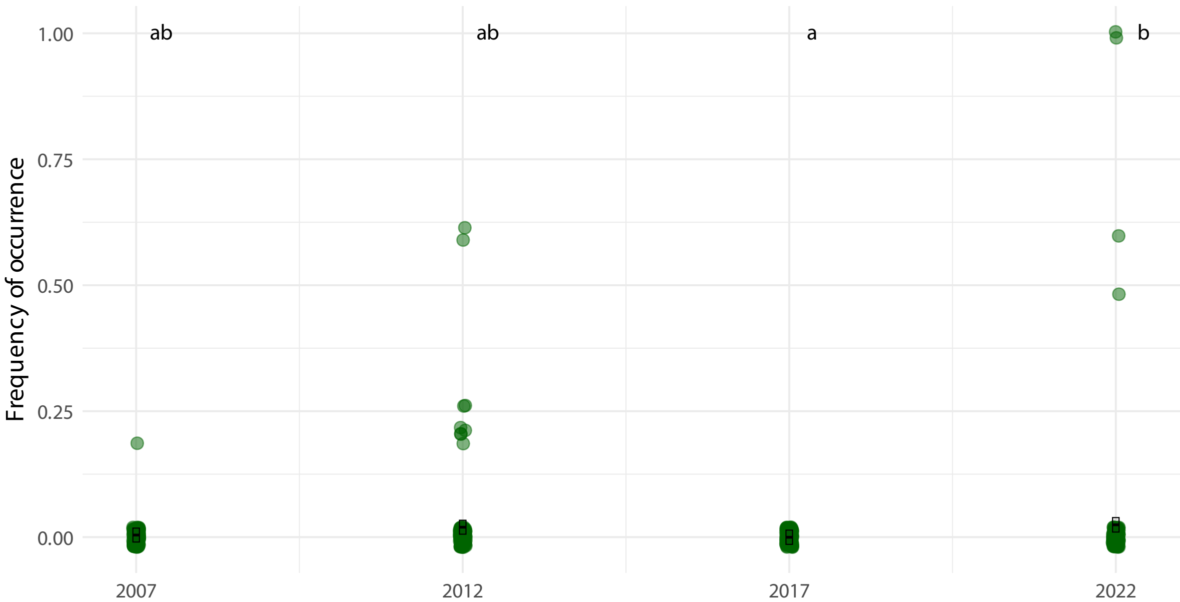

Figure 18: Frequency of occurrence for *Agrostis mertensii* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

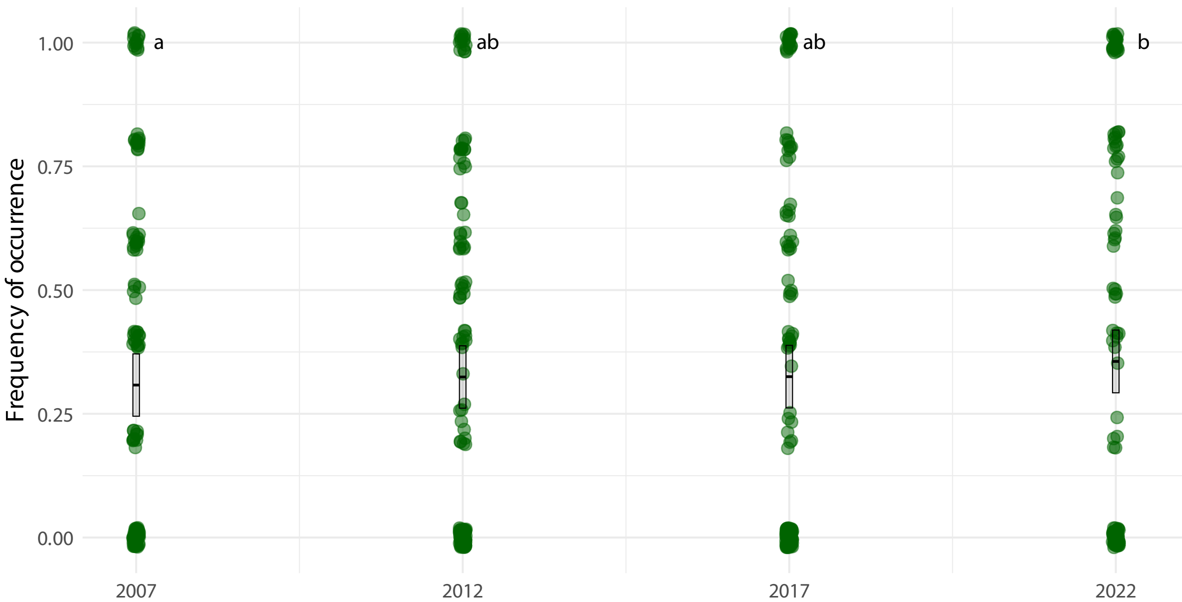

Figure 19: Frequency of occurrence for *Betula nana* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

Figure 20: Frequency of occurrence for *Bryophyte* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

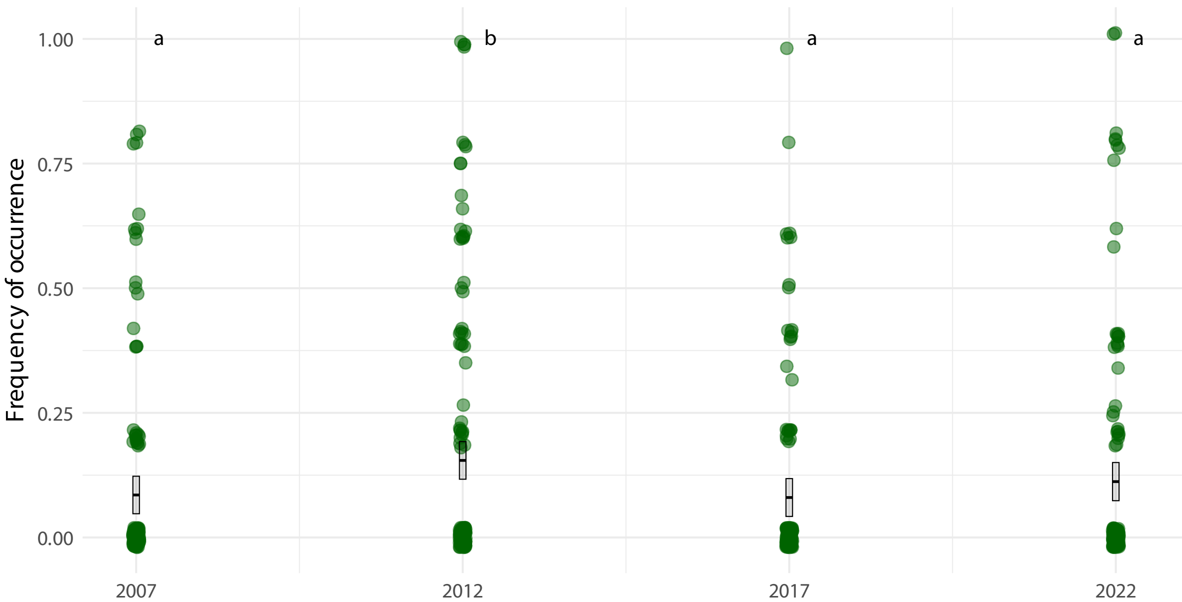

Figure 21: Frequency of occurrence for *Coptis trifolia* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

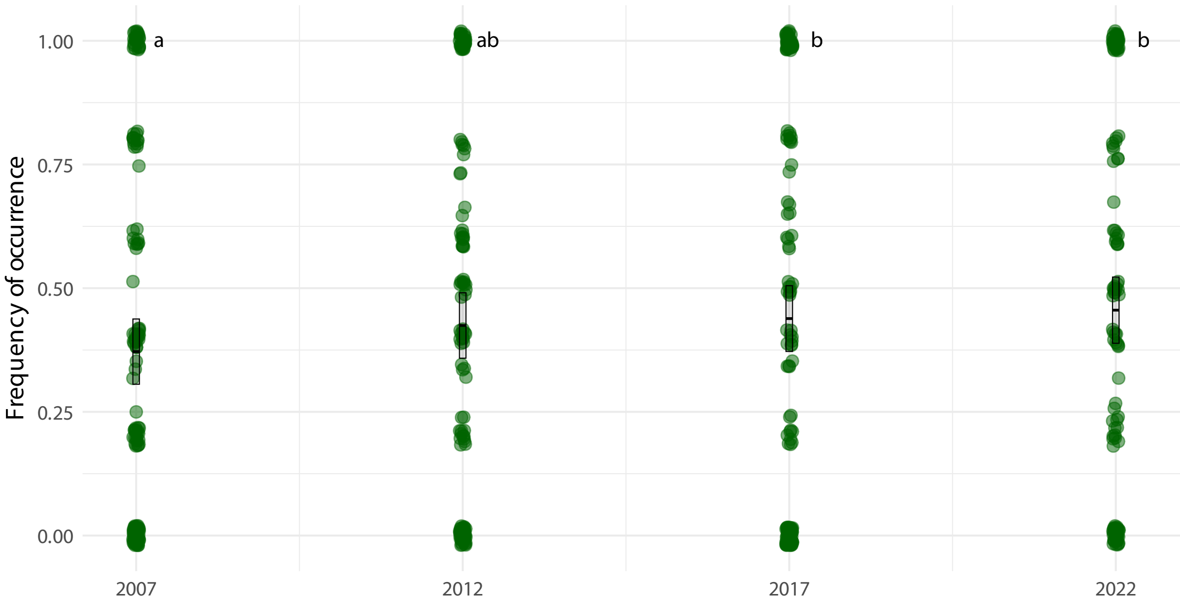

Figure 22: Frequency of occurrence for *Deschampsia flexuosa* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

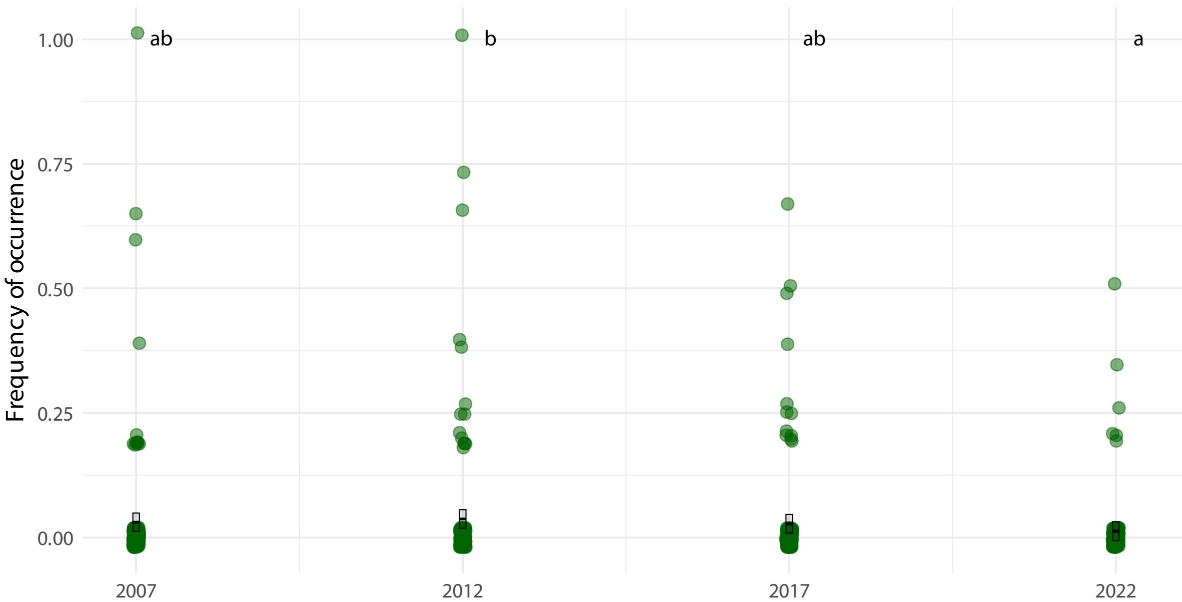

Figure 23: Frequency of occurrence for *Diphasiastrum alpinum* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

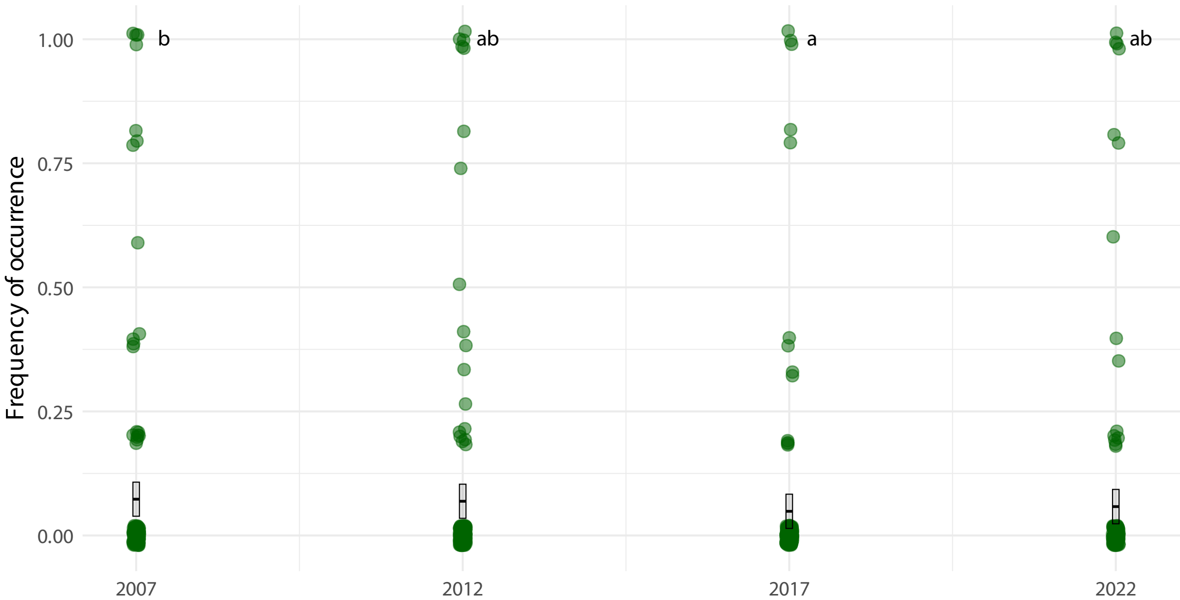

Figure 24: Frequency of occurrence for *Eriophorum angustifolium* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

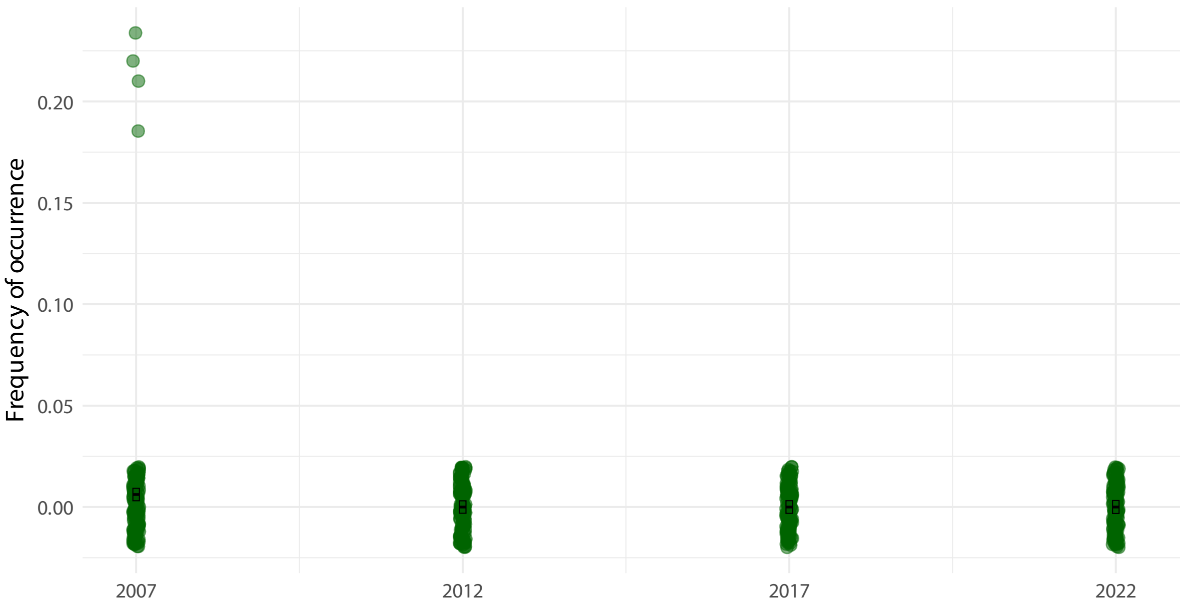

Figure 25: Frequency of occurrence for *Hieracium sp* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

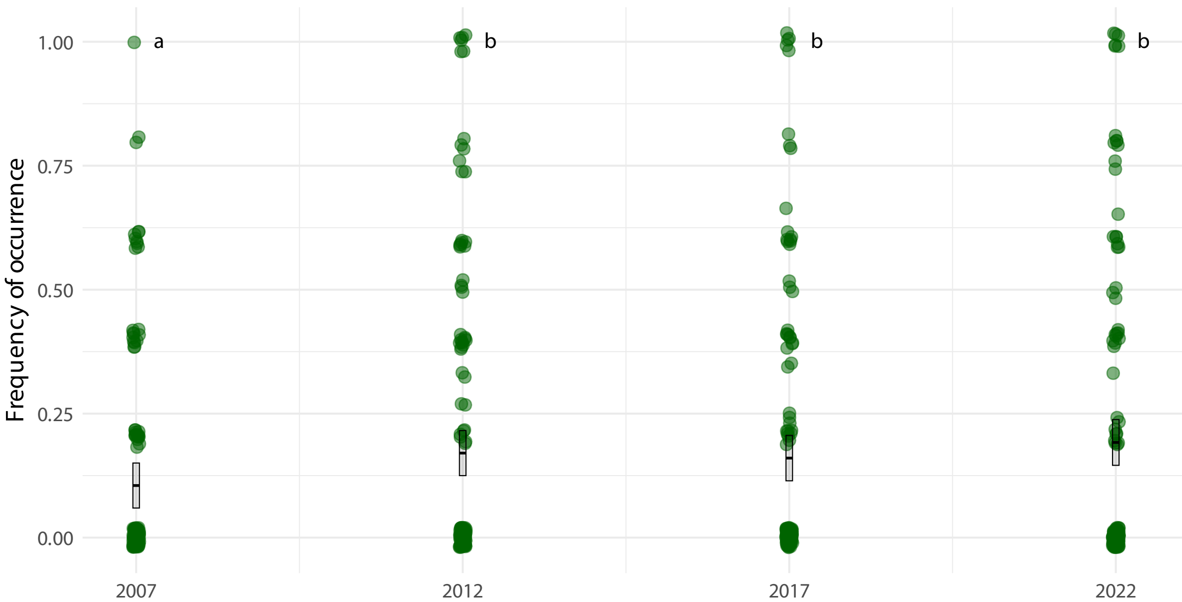

Figure 26: Frequency of occurrence for *Ledum groenlandicum* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

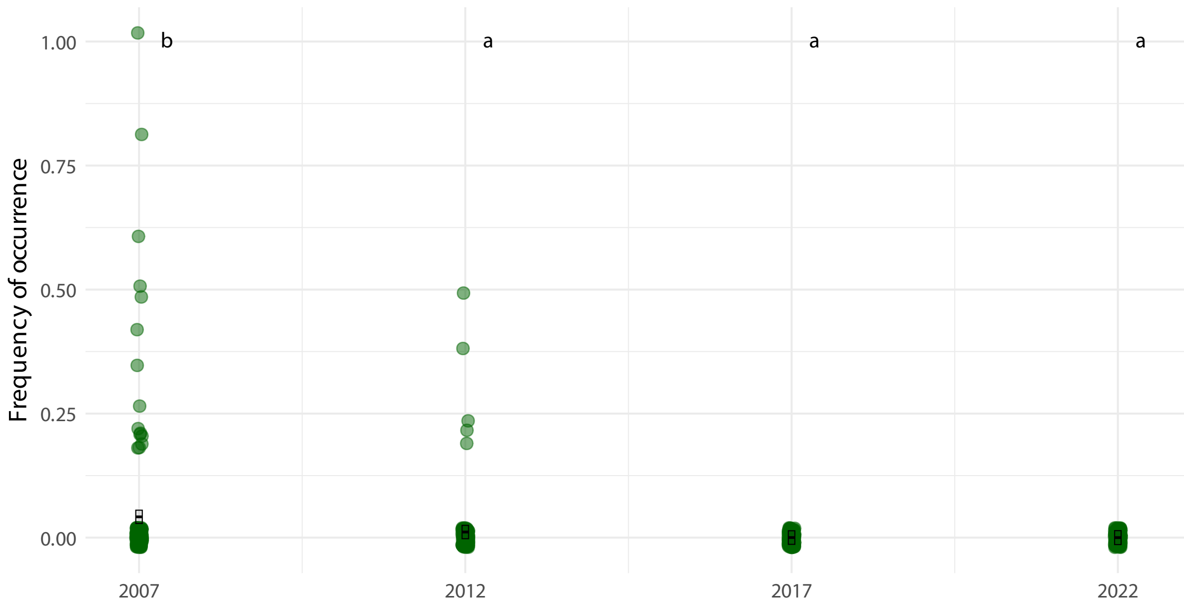

Figure 27: Frequency of occurrence for *Ledum palustre* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

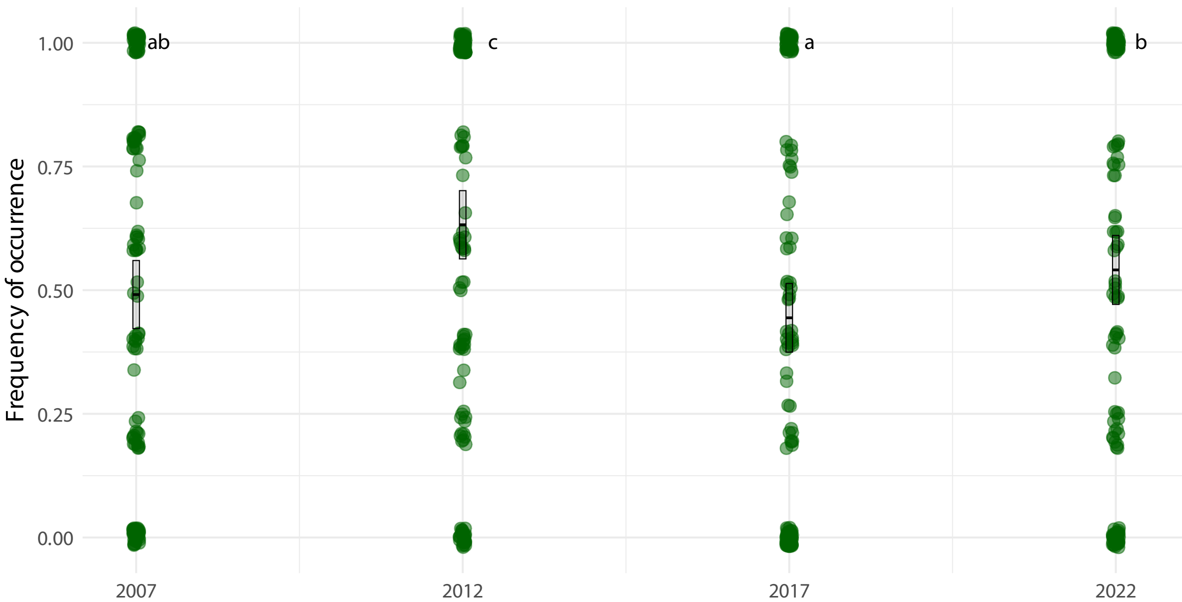

Figure 28: Frequency of occurrence for *Lichen* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

Figure 29: Frequency of occurrence for *Lycopodium annotinum* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

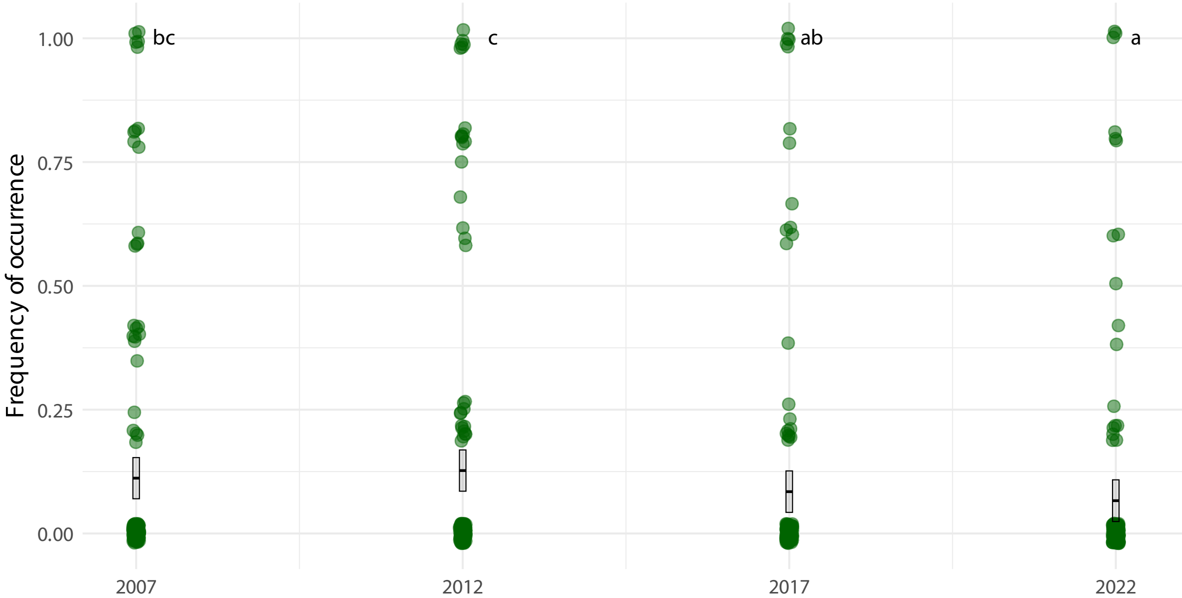

Figure 30: Frequency of occurrence for *Poa pratensis* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

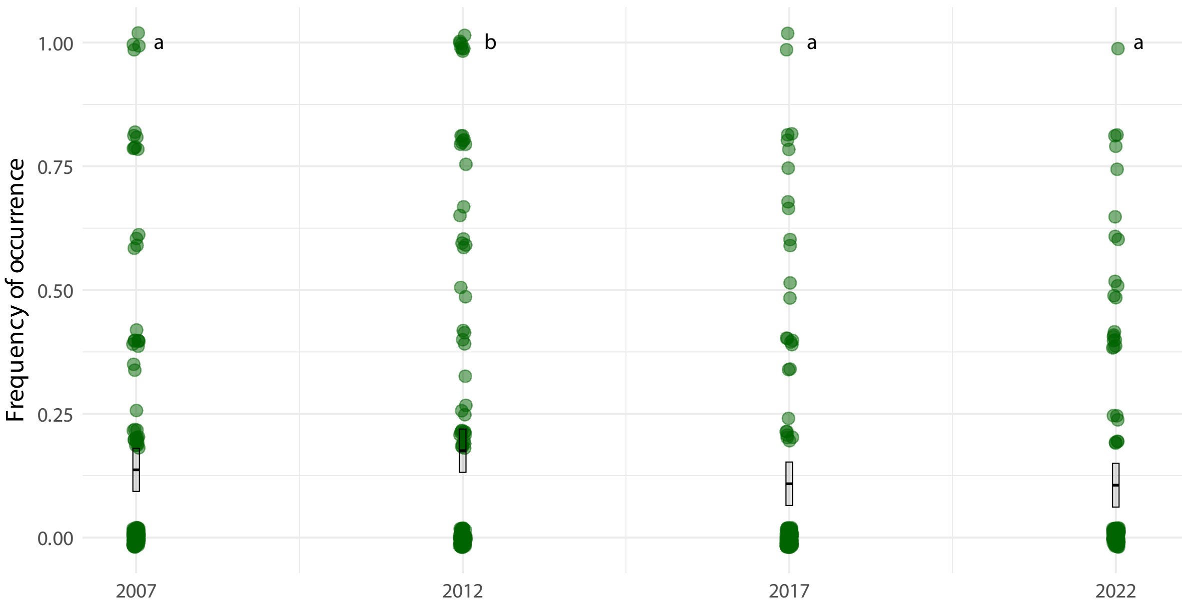

Figure 31: Frequency of occurrence for *Polygonum vivipara* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

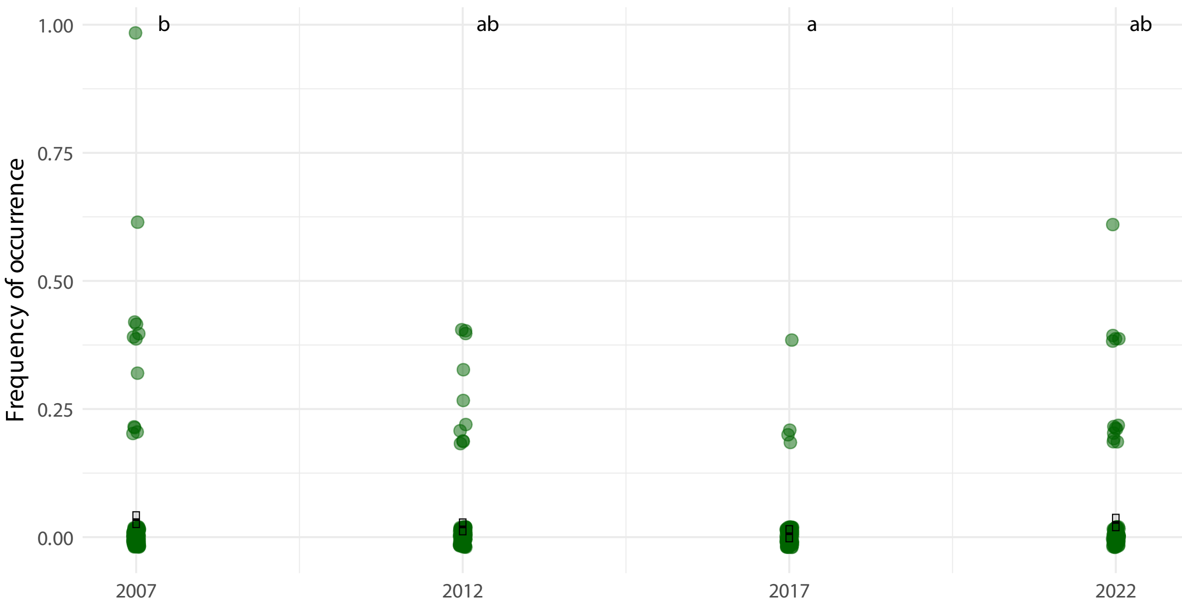

Figure 32: Frequency of occurrence for *Pyrola minor* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

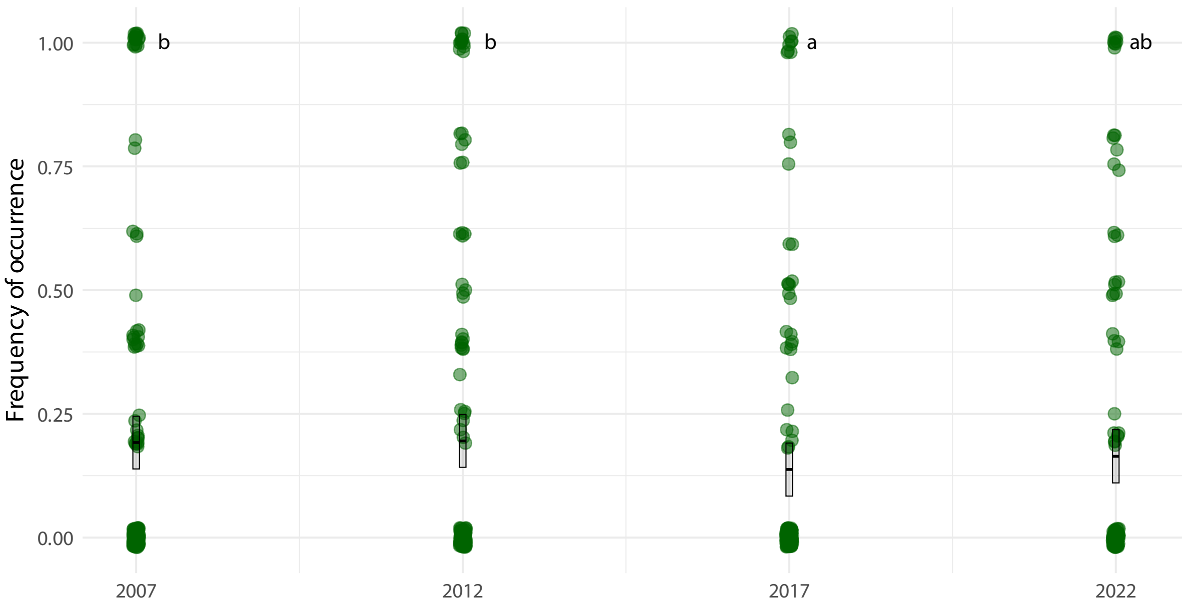

Figure 33: Frequency of occurrence for *Salix herbacea* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

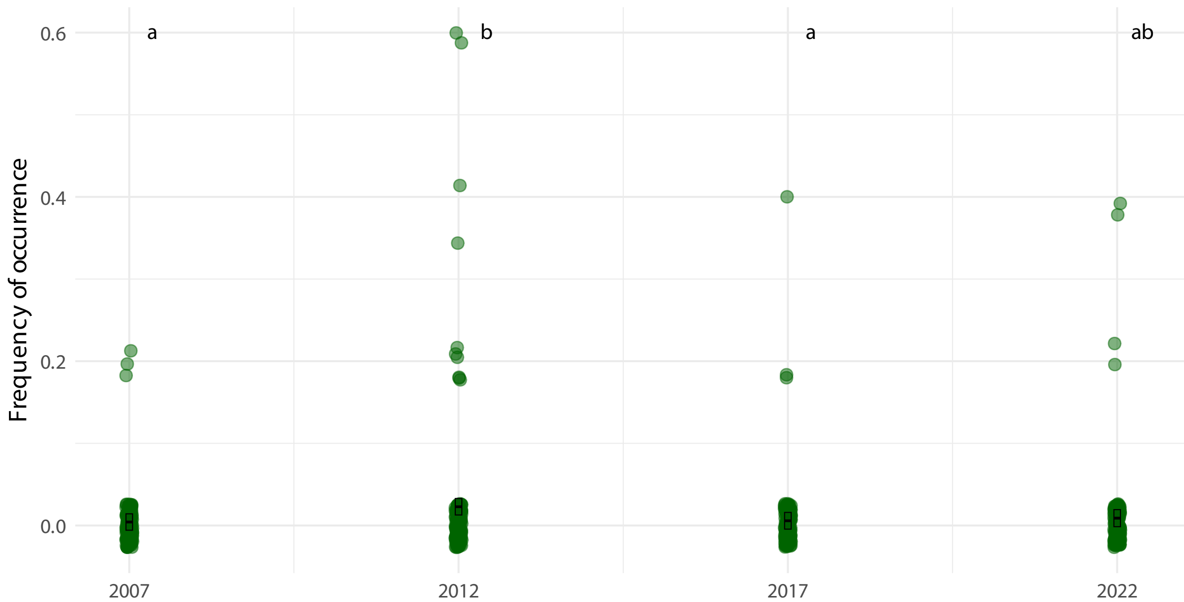

Figure 34: Frequency of occurrence for *Stellaria longipes* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

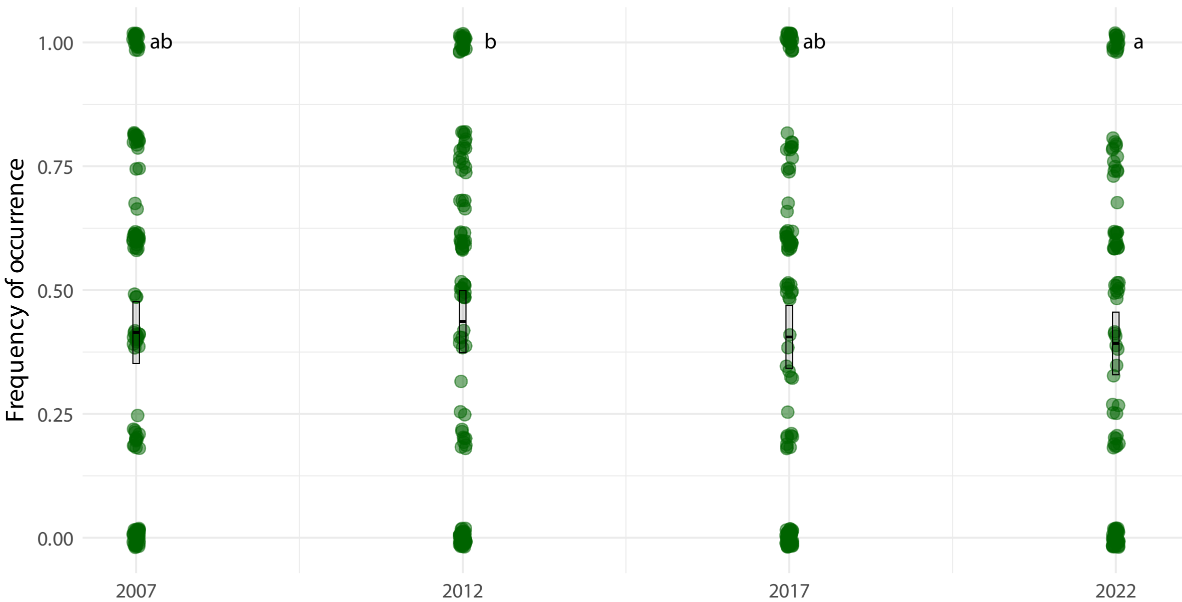

Figure 35: Frequency of occurrence for *Vaccinium uliginosum* across survey years. Points show raw observations, crossbars show estimated marginal means ± 95% CI. Compact letter display indicates statistically significant differences between years (Holm adjustment).

4 Phenology (discussion)

4.1 NDVI

Data available from https://data.g-e-m.dk (GEM 2026b).

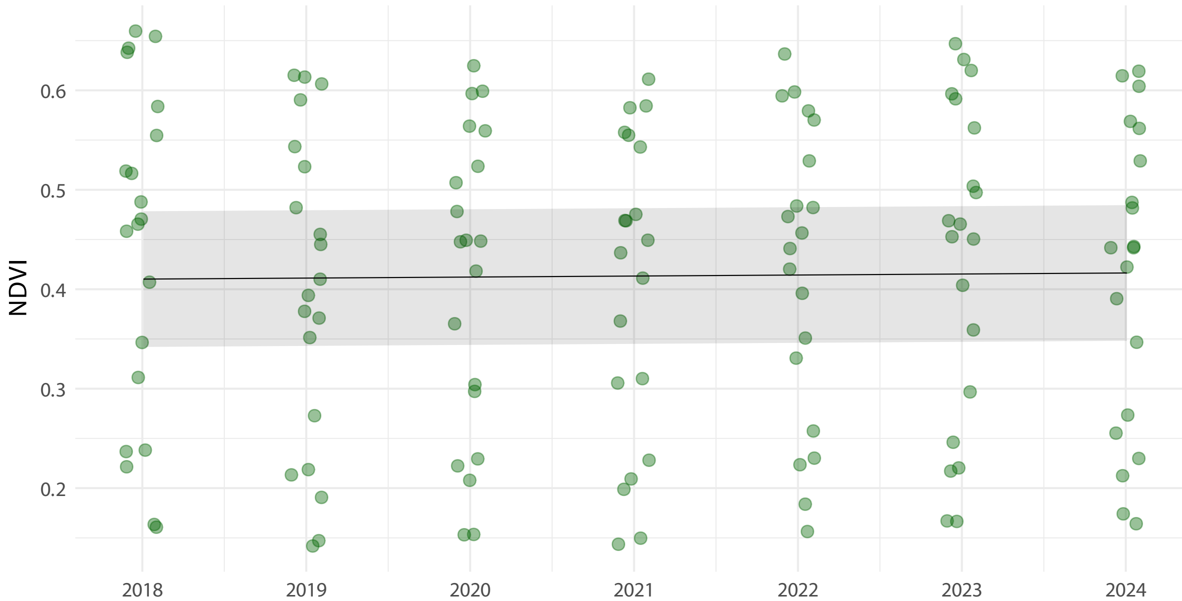

Figure 36: NDVI change over time. Mean NDVI per plot per year, based on in situ measurements from 2018 to 2024. The trend line shows a significant increase of 0.001 NDVI units per year (95% CI: 3e-04–0.0018), representing a cumulative increase of 0.0061 NDVI units over the study period.

References

Cappelen, John. 2021. *Greenland – DMI Historical Climate Data Collection 1784-2020*. Nos. 21–04. Danish Meteorological Institute. https://www.dmi.dk/fileadmin/Rapporter/2021/DMIRep21-04.pdf.

GEM. 2026a. "BioBasis Nuuk - Vegetation - NERO Line Raunkiaer." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/HXJY-RQ13.

GEM. 2026b. "BioBasis Nuuk - Vegetation - Plot NDVI." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/TQTT-EV69.

GEM. 2026c. "ClimateBasis Nuuk - Precipitation - Precipitation Accumulated (Mm)." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/SXJ8-WA79.

GEM. 2026d. "ClimateBasis Nuuk - Radiation - Photosyntetic Active Radiation @ 200 Cm - 5min Average (Mmol/M2/Sec)." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/8Z2W-D993.

GEM. 2026e. "ClimateBasis Nuuk - Temperature - Air Temperature @ 200 Cm - 30min Average (°C)." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/PGN3-7597.